INCH-POUND

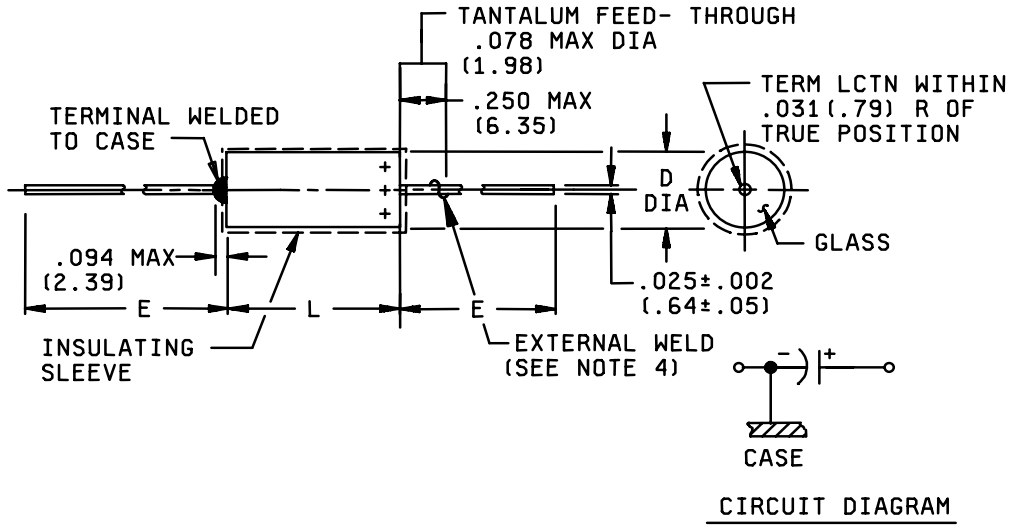
MIL-PRF-39006/30B
20 October 2005
SUPERSEDING
MIL-PRF-39006/30A
17 April 2002

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (POLARIZED, SINTERED SLUG), 85°C (VOLTAGE DERATED TO 125°C), ESTABLISHED RELIABILITY, STYLE CLR90

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-39006.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. The weld shall not be enclosed in the end seal.

FIGURE 1. Style CLR90 capacitors.

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Case size	Dimensions			
	Basic case		Insulated case	
	L 1/ +.031 (0.79) -.016 (0.41)	D ±.016 (0.41)	D max	E ±.250 (6.35)
T1	.453 (11.51)	.188 (4.78)	.219 (5.56)	1.500 (38.10)
T2	.641 (16.28)	.281 (7.14)	.312 (7.92)	2.250 (57.15)
T3	.766 (19.46)	.375 (9.52)	.406 (10.31)	2.250 (57.15)
T4	1.062 (26.97)	.375 (9.52)	.406 (10.31)	2.250 (57.15)

1/ Length of basic case sleeving shall be as specified in MIL-PRF-39006.

FIGURE 1. Style CLR90 capacitors - Continued.

QUALIFICATION:

A manufacturer shall be considered qualified to this specification sheet by similarity if the manufacturer is qualified to style CLR79 capacitors in MIL-PRF-39006/22 and if the design and construction of the capacitors in this specification sheet is the same as the design and construction of the capacitors covered by MIL-PRF-39006/22.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Tantalum, tubular, insulated.

Seal type: Hermetic (glass-to-tantalum anode). For compression type, see table I.

Terminals: Axial-wire lead: Cathode and anode (N32, N51, or N52 of MIL-STD-1276).

Rated temperature: -55°C to +85°C, voltage derated to +125°C (see figure 2).

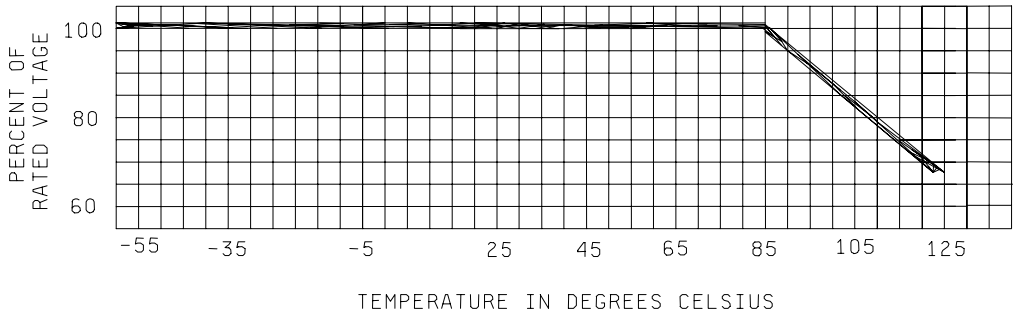


FIGURE 2. Voltage derating with temperature.

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TABLE I. Capacitor characteristics.

Rated voltage (+85°C) 1/ V dc	Cap. µF	Cap. tolerance percent	DC leakage (max) +25°C µA	DC leakage (max) +85°C, +125°C µA	Dissipation factor (max) percent	ESR at +25°C (max) 2/ Ohms	Derated voltage (+125°C) 1/ V dc	Surge voltage (+85°C) V dc	Impedance (max) Ohms	Capacitance change at -55°C percent	Capacitance change at +85°C percent	Capacitance change at +125°C percent	Max 85°C 40 kHz ripple current 3/ mA rms	Case size	PIN M39006/30- 4/ 5/ noncompression seal Failure rate level (%/1,000 hour) M(1.0)	P(0.1)	R(0.01)	S(0.001)	PIN M39006/30- 4/ 5/ compression seal Failure rate level (%/1,000 hour) M(1.0)	P(0.1)	R(0.01)	S(0.001)
6	30	±20	1	2	4.5	1.99	4	6.9	100	-40	+10.5	+12	820	T1	0001-	0221-	0441-	0661-	1001-	2011-	3041-	4061-
6	30	±10	1	2	4.5	1.99	4	6.9	100	-40	+10.5	+12	820	T1	0002-	0222-	0442-	0662-	1002-	2012-	3042-	4062-
6	30	±5	1	2	4.5	1.99	4	6.9	100	-40	+10.5	+12	820	T1	0003-	0223-	0443-	0663-	1003-	2013-	3043-	4063-
6	68	±20	1	2	7.5	1.58	4	6.9	60	-40	+14	+16	960	T1	0004-	0224-	0444-	0664-	1004-	2014-	3044-	4064-
6	68	±10	1	2	7.5	1.58	4	6.9	60	-40	+14	+16	960	T1	0005-	0225-	0445-	0665-	1005-	2015-	3045-	4065-
6	68	±5	1	2	7.5	1.58	4	6.9	60	-40	+14	+16	960	T1	0006-	0226-	0446-	0666-	1006-	2016-	3046-	4066-
6	140	±20	1	3	10.5	0.99	4	6.9	40	-40	+14	+16	1,200	T2	0007-	0227-	0447-	0667-	1007-	2017-	3047-	4067-
6	140	±10	1	3	10.5	0.99	4	6.9	40	-40	+14	+16	1,200	T2	0008-	0228-	0448-	0668-	1008-	2018-	3048-	4068-
6	140	±5	1	3	10.5	0.99	4	6.9	40	-40	+14	+16	1,200	T2	0009-	0229-	0449-	0669-	1009-	2019-	3049-	4069-
6	270	±20	1	6.5	22.5	1.11	4	6.9	25	-44	+17.5	+20	1,375	T2	0010-	0230-	0450-	0670-	1010-	2020-	3050-	4070-
6	270	±10	1	6.5	22.5	1.11	4	6.9	25	-44	+17.5	+20	1,375	T2	0011-	0231-	0451-	0671-	1011-	2021-	3051-	4071-
6	270	±5	1	6.5	22.5	1.11	4	6.9	25	-44	+17.5	+20	1,375	T2	0012-	0232-	0452-	0672-	1012-	2022-	3052-	4072-
6	330	±20	2	7.9	18	0.73	4	6.9	20	-44	+14	+16	1,800	T3	0013-	0233-	0453-	0673-	1013-	2023-	3053-	4073-
6	330	±10	2	7.9	18	0.73	4	6.9	20	-44	+14	+16	1,800	T3	0014-	0234-	0454-	0674-	1014-	2024-	3054-	4074-
6	330	±5	2	7.9	18	0.73	4	6.9	20	-44	+14	+16	1,800	T3	0015-	0235-	0455-	0675-	1015-	2025-	3055-	4075-
6	560	±20	2	13	27.5	0.65	4	6.9	25	-64	+17.5	+20	1,900	T3	0016-	0236-	0456-	0676-	1016-	2026-	3056-	4076-
6	560	±10	2	13	27.5	0.65	4	6.9	25	-64	+17.5	+20	1,900	T3	0017-	0237-	0457-	0677-	1017-	2027-	3057-	4077-
6	560	±5	2	13	27.5	0.65	4	6.9	25	-64	+17.5	+20	1,900	T3	0018-	0238-	0458-	0678-	1018-	2028-	3058-	4078-
6	1,200	±20	3	14	45	0.50	4	6.9	20	-80	+25	+25	2,265	T4	0019-	0239-	0459-	-0679-	1019-	2029-	3059-	4079-
6	1,200	±10	3	14	45	0.50	4	6.9	20	-80	+25	+25	2,265	T4	0020-	0240-	0460-	0680-	1020-	2030-	3060-	4080-
8	25	±20	1	2	3.75	1.99	5	9.2	100	-40	+10.5	+12	820	T1	0021-	0241-	0461-	0681-	1021-	2031-	3061-	4081-
8	25	±10	1	2	3.75	1.99	5	9.2	100	-40	+10.5	+12	820	T1	0022-	0242-	0462-	0682-	1022-	2032-	3062-	4082-
8	25	±5	1	2	3.75	1.99	5	9.2	100	-40	+10.5	+12	820	T1	0023-	0243-	0463-	0683-	1023-	2033-	3063-	4083-
8	56	±20	1	2	7	1.66	5	9.2	59	-40	+14	+16	900	T1	0024-	0244-	0464-	0684-	1024-	2034-	3064-	4084-
8	56	±10	1	2	7	1.66	5	9.2	59	-40	+14	+16	900	T1	0025-	0245-	0465-	0685-	1025-	2035-	3065-	4085-
8	56	±5	1	2	7	1.66	5	9.2	59	-40	+14	+16	900	T1	0026-	0246-	0466-	0686-	1026-	2036-	3066-	4086-
8	120	±20	1	2	10	1.11	5	9.2	50	-44	+17.5	+20	1,220	T2	0027-	0247-	0467-	0687-	1027-	2037-	3067-	4087-
8	120	±10	1	2	10	1.11	5	9.2	50	-44	+17.5	+20	1,220	T2	0028-	0248-	0468-	0688-	1028-	2038-	3068-	4088-
8	120	±5	1	2	10	1.11	5	9.2	50	-44	+17.5	+20	1,220	T2	0029-	0249-	0469-	0689-	1029-	2039-	3069-	4089-
8	220	±20	1	7	18.5	1.12	5	9.2	30	-44	+17.5	+20	1,370	T2	0030-	0250-	0470-	0690-	1030-	2040-	3070-	4090-
8	220	±10	1	7	18.5	1.12	5	9.2	30	-44	+17.5	+20	1,370	T2	0031-	0251-	0471-	0691-	1031-	2041-	3071-	4091-
8	220	±5	1	7	18.5	1.12	5	9.2	30	-44	+17.5	+20	1,370	T2	0032-	0252-	0472-	0692-	1032-	2042-	3072-	4092-
8	290	±20	2	6	17	0.78	5	9.2	25	-64	+17.5	+20	1,770	T3	0033-	0253-	0473-	0693-	1033-	2043-	3073-	4093-
8	290	±10	2	6	17	0.78	5	9.2	25	-64	+17.5	+20	1,770	T3	0034-	0254-	0474-	0694-	1034-	2044-	3074-	4094-
8	290	±5	2	6	17	0.78	5	9.2	25	-64	+17.5	+20	1,770	T3	0035-	0255-	0475-	0695-	1035-	2045-	3075-	4095-
8	430	±20	2	14	23	0.71	5	9.2	25	-64	+17.5	+20	1,825	T3	0036-	0256-	0476-	0696-	1036-	2046-	3076-	4096-

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

Rated voltage (+85°C) 1/ V dc	Cap. µF	Cap. tolerance percent	DC leakage (max) +25°C µA	DC leakage (max) +85°C, +125°C µA	Dissipation factor (max) percent	ESR at +25°C (max) 2/ Ohms	Derated voltage (+125°C) 1/ V dc	Surge voltage (+85°C) V dc	Impedance (max) Ohms	Capacitance change at -55°C percent	Capacitance change at +85°C percent	Capacitance change at +125°C percent	Max 85°C 40 kHz ripple current 3/ mA rms	Case size	PIN M39006/30- 4/ 5/ noncompression seal Failure rate level (%/1,000 hour) M(1.0)	P(0.1)	R(0.01)	S(0.001)	PIN M39006/30- 4/ 5/ compression seal Failure rate level (%/1,000 hour) M(1.0)	P(0.1)	R(0.01)	S(0.001)
8	430	±10	2	14	23	0.71	5	9.2	25	-64	+17.5	+20	1,825	T3	0037-	0257-	0477-	0697-	1037-	2047-	3077-	4097-
8	430	±5	2	14	23	0.71	5	9.2	25	-64	+17.5	+20	1,825	T3	0038-	0258-	0478-	0698-	1038-	2048-	3078-	4098-
8	850	±20	4	16	30	0.47	5	9.2	22	-80	+25	+25	2,330	T4	0039-	0259-	0479-	0699-	1039-	2049-	3079-	4099-
8	850	±10	4	16	30	0.47	5	9.2	22	-80	+25	+25	2,330	T4	0040-	0260-	0480-	0700-	1040-	2050-	3080-	4100-
10	20	±20	1	2	3	1.99	7	11.5	175	-32	+10.5	+12	820	T1	0041-	0261-	0481-	0701-	1041-	2051-	3081-	4101-
10	20	±10	1	2	3	1.99	7	11.5	175	-32	+10.5	+12	820	T1	0042-	0262-	0482-	0702-	1042-	2052-	3082-	4102-
10	20	±5	1	2	.3	1.99	7	11.5	175	-32	+10.5	+12	820	T1	0043-	0263-	0483-	0703-	1043-	2053-	3083-	4103-
10	47	±20	1	2	6.5	1.84	7	11.5	100	-36	+14	+16	855	T1	0044-	0264-	0484-	0704-	1044-	2054-	3084-	4104-
10	47	±10	1	2	6.5	1.84	7	11.5	100	-36	+14	+16	855	T1	0045-	0265-	0485-	0705-	1045-	2055-	3085-	4105-
10	47	±5	1	2	6.5	1.84	7	11.5	100	-36	+14	+16	855	T1	0046-	0266-	0486-	0706-	1046-	2056-	3086-	4106-
10	100	±20	1	4	7.5	0.99	7	11.5	60	-36	+14	+16	1,200	T2	0047-	0267-	0487-	0707-	1047-	2057-	3087-	4107-
10	100	±10	1	4	7.5	0.99	7	11.5	60	-36	+14	+16	1.200	T2	0048-	0268-	0488-	0708-	1048-	2058-	3088-	4108-
10	100	±5	1	4	7.5	0.99	7	11.5	60	-36	+14	+16	1,200	T2	0049-	0269-	0489-	0709-	1049-	2059-	3089-	4109-
10	180	±20	1	7	15	1.11	7	11.5	40	-36	+14	+16	1,365	T2	0050-	0270-	0490-	0710-	1050-	2060-	3090-	4110-
10	180	±10	1	7	15	1.11	7	11.5	40	-36	+14	+16	1,365	T2	0051-	0271-	0491-	0711-	1051-	2061-	3091-	4111-
10	180	±5	1	7	15	1.11	7	11.5	40	-36	+14	+16	1,365	T2	0052-	0272-	0492-	0712-	1052-	2062-	3092-	4112-
10	250	±20	2	10	15	0.80	7	11.5	30	-40	+14	+16	1,720	T3	0053-	0273-	0493-	0713-	1053-	2063-	3093-	4113-
10	250	±10	2	10	15	0.80	7	11.5	30	-40	+14	+16	1,720	T3	0054-	0274-	0494-	0714-	1054-	2064-	3094-	4114-
10	250	±5	2	10	15	0.80	7	11.5	30	-40	+14	+16	1,720	T3	0055-	0275-	0495-	-0715-	1055-	2065-	3095-	4115-
10	390	±20	2	16	22	0.75	7	11.5	25	-64	+17.5	+20	1,800	T3	0056-	0276-	0496-	0716-	1056-	2066-	3096-	4116-
10	390	±10	2	16	22	0.75	7	11.5	25	-64	+17.5	+20	1,800	T3	0057-	0277-	0497-	0717-	1057-	2067-	3097-	4117-
10	390	±5	2	16	22	0.75	7	11.5	25	-64	+17.5	+20	1,800	T3	0058-	0278-	0498-	0718-	1058-	2068-	3098-	4118-
10	750	±20	4	16	25	0.44	7	11.5	23	-80	+25	+25	2.360	T4	0059-	0279-	0499-	0719-	1059-	2069-	3099-	4119-
10	750	±10	4	16	25	0.44	7	11.5	23	-80	+25	+25	2,360	T4	0060-	0280-	0500-	0720-	1060-	2070-	3100-	4120-
15	15	±20	1	2	2.5	2.21	10	17.2	155	-24	+10.5	+12	780	T1	0061-	0281-	0501-	0721-	1061-	2071-	3101-	4121-
15	15	±10	1	2	2.5	2.21	10	17.2	155	-24	+10.5	+12	780	T1	0062-	0282-	0502-	0722-	1062-	2072-	3102-	4122-
15	15	±5	1	2	2.5	2.21	10	17.2	155	-24	+10.5	+12	780	T1	0063-	0283-	0503-	0723-	1063-	2073-	3103-	4123-
15	33	±20	1	2	5	2.01	10	17.2	90	-28	+14	+16	820	T1	0064-	0284-	0504-	0724-	1064-	2074-	3104-	4124-
15	33	±10	1	2	5	2.01	10	17.2	90	-28	+14	+16	820	T1	0065-	0285-	0505-	0725-	1065-	2075-	3105-	4125-
15	33	±5	1	2	5	2.01	10	17.2	90	-28	+14	+16	820	T1	0066-	0286-	0506-	0726-	1066-	2076-	3106-	4126-
15	70	±20	1	4	6.5	1.23	10	17.2	75	-28	+14	+16	1,150	T2	0067-	0287-	0507-	0727-	1067-	2077-	3107-	4127-
15	70	±10	1	4	6.5	1.23	10	17.2	75	-28	+14	+16	1,150	T2	0068-	0288-	0508-	0728-	1068-	2078-	3108-	4128-
15	70	±5	1	4	6.5	1.23	10	17.2	75	-28	+14	+16	1,150	T2	0069-	0289-	0509-	0729-	1069-	2079-	3109-	4129-
15	120	±20	1	7	9	0.99	10	17.2	50	-28	+17.5	+20	1,450	T2	0070-	0290-	0510-	0730-	1070-	2080-	3110-	4130-
15	120	±10	1	7	9	0.99	10	17.2	50	-28	+17.5	+20	1,450	T2	0071-	0291-	0511-	0731-	1071-	2081-	3111-	4131-
15	120	±5	1	7	9	0.99	10	17.2	50	-28	+17.5	+20	1,450	T2	0072-	0292-	0512-	0732-	1072-	2082-	3112-	4132-

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

Rated voltage (+85°C) 1/ V dc	Cap. µF	Cap. tolerance percent	DC leakage (max) +25°C µA	DC leakage (max) +85°C, +125°C µA	Dissipation factor (max) percent	ESR at +25°C (max) 2/ Ohms	Derated voltage (+125°C) 1/ V dc	Surge voltage (+85°C) V dc	Impedance (max) Ohms	Capacitance change at -55°C percent	Capacitance change at +85°C percent	Capacitance change at +125°C percent	Max 85°C 40 kHz ripple current 3/ mA rms	Case size	PIN M39006/30- 4/ 5/ noncompression seal Failure rate level (%/1,000 hour) M(1.0)	P(0.1)	R(0.01)	S(0.001)	PIN M39006/30- 4/ 5/ compression seal Failure rate level (%/1,000 hour) M(1.0)	P(0.1)	R(0.01)	S(0.001)
15	170	±20	2	10	12.5	0.98	10	17.2	35	-32	+14	+16	1,480	T3	0073-	0293-	0513-	0733-	1073-	2083-	3113-	4133-
15	170	±10	2	10	12.5	0.98	10	17.2	35	-32	+14	+16	1,480	T3	0074-	0294-	0514-	0734-	1074-	2084-	3114-	4134-
15	170	± 5	2	10	12.5	0.98	10	17.2	35	-32	+14	+16	1,480	T3	0075-	0295-	0515-	0735-	1075-	2085-	3115-	4135-
15	270	±20	2	16	16	0.79	10	17.2	30	-56	+17.5	+20	1,740	T3	0076-	0296-	0516-	0736-	1076-	2086-	3116-	4136-
15	270	±10	2	16	16	0.79	10	17.2	30	-56	+17.5	+20	1,740	T3	0077-	0297-	0517-	0737-	1077-	2087-	3117-	4137-
15	270	± 5	2	16	16	0.79	10	17.2	30	-56	+17.5	+20	1,740	T3	0078-	0298-	0518-	0738-	1078-	2088-	3118-	4138-
15	540	±20	6	24	20	0.49	10	17.2	23	-80	+25	+25	2,330	T4	0079-	0299-	0519-	0739-	1079-	2089-	3119-	4139-
15	540	±10	6	24	20	0.49	10	17.2	23	-80	+25	+25	2,330	T4	0080-	0300-	0520-	0740-	1080-	2090-	3120-	4140-
25	10	±20	1	2	2	2.66	15	28.8	220	-16	+ 8	+ 9	715	T1	0081-	0301-	0521-	0741-	1081-	2091-	3121-	4141-
25	10	±10	1	2	2	2.66	15	28.8	220	-16	+ 8	+ 9	715	T1	0082-	0302-	0522-	0742-	1082-	2092-	3122-	4142-
25	10	± 5	1	2	2	2.66	15	28.8	220	-16	+ 8	+ 9	715	T1	0083-	0303-	0523-	0743-	1083-	2093-	3123-	4143-
25	22	±20	1	2	3.3	1.99	15	28.8	140	-20	+10.5	+12	825	T1	0084-	0304-	0524-	0744-	1084-	2094-	3124-	4144-
25	22	±10	1	2	3.3	1.99	15	28.8	140	-20	+10.5	+12	825	T1	0085-	0305-	0525-	0745-	1085-	2095-	3125-	4145-
25	22	± 5	1	2	3.3	1.99	15	28.8	140	-20	+10.5	+12	825	T1	0086-	0306-	0526-	0746-	1086-	2096-	3126-	4146-
25	50	±20	1	2	5.5	1.46	15	28.8	70	-28	+13	+15	1,130	T2	0087-	0307-	0527-	0747-	1087-	2097-	3127-	4147-
25	50	±10	1	2	5.5	1.46	15	28.8	70	-28	+13	+15	1,130	T2	0088-	0308-	0528-	0748-	1088-	2098-	3128-	4148-
25	50	± 5	1	2	5.5	1.46	15	28.8	70	-28	+13	+15	1,130	T2	0089-	0309-	0529-	0749-	1089-	2099-	3129-	4149-
25	100	±20	1	10	7.5	0.99	15	28.8	50	-28	+13	+15	1,435	T2	0090-	0310-	0530-	0750-	1090-	2100-	3130-	4150-
25	100	±10	1	10	7.5	0.99	15	28.8	50	-28	+13	+15	1,435	T2	0091-	0311-	0531-	0751-	1091-	2101-	3131-	4151-
25	100	± 5	1	10	7.5	0.99	15	28.8	50	-28	+13	+15	1,435	T2	0092-	0312-	0532-	0752-	1092-	2102-	3132-	4152-
25	120	±20	2	6	10.5	1.16	15	28.8	38	-32	+13	+15	1,450	T3	0093-	0313-	0533-	0753-	1093-	2103-	3133-	4153-
25	120	±10	2	6	10.5	1.16	15	28.8	38	-32	+13	+15	1,450	T3	0094-	0314-	0534-	0754-	1094-	2104-	3134-	4154-
25	120	± 5	2	6	10.5	1.16	15	28.8	38	-32	+13	+15	1,450	T3	0095-	0315-	0535-	0755-	1095-	2105-	3135-	4155-
25	180	±20	2	18	13	0.96	15	28.8	32	-48	+13	+15	1,525	T3	0096-	0316-	0536-	0756-	1096-	2106-	3136-	4156-
25	180	±10	2	18	13	0.96	15	28.8	32	-48	+13	+15	1,525	T3	0097-	0317-	0537-	0757-	1097-	2107-	3137-	4157-
25	180	± 5	2	18	13	0.96	15	28.8	32	-48	+13	+15	1,525	T3	0098-	0318-	0538-	0758-	1098-	2108-	3138-	4158-
25	350	±20	7	28	17.5	0.67	15	28.8	24	-70	+25	+25	1,970	T4	0099-	0319-	0539-	0759-	1099-	2109-	3139-	4159-
25	350	±10	7	28	17.5	0.67	15	28.8	24	-70	+25	+25	1,970	T4	0100-	0320-	0540-	0760-	1100-	2110-	3140-	4160-
30	8	±20	1	2	2	3.32	20	34.5	275	-16	+ 8	+12	640	T1	0101-	0321-	0541-	0761-	1101-	2111-	3141-	4161-
30	8	±10	1	2	2	3.32	20	34.5	275	-16	+ 8	+12	640	T1	0102-	0322-	0542-	0762-	1102-	2112-	3142-	4162-
30	8	± 5	1	2	2	3.32	20	34.5	275	-16	+ 8	+12	640	T1	0103-	0323-	0543-	0763-	1103-	2113-	3143-	4163-
30	15	±20	1	2	2.5	2.21	20	34.5	175	-20	+10.5	+12	780	T1	0104-	0324-	0544-	0764-	1104-	2114-	3144-	4164-
30	15	±10	1	2	2.5	2.21	20	34.5	175	-20	+10.5	+12	780	T1	0105-	0325-	0545-	0765-	1105-	2115-	3145-	4165-
30	15	± 5	1	2	2.5	2.21	20	34.5	175	-20	+10.5	+12	780	T1	0106-	0326-	0546-	0766-	1106-	2116-	3146-	4166-
30	40	±20	1	5	5	1.66	20	34.5	65	-24	+10.5	+12	1,120	T2	0107-	0327-	0547-	0767-	1107-	2117-	3147-	4167-
30	40	±10	1	5	5	1.66	20	34.5	65	-24	+10.5	+12	1,120	T2	0108-	0328-	0548-	0768-	1108-	2118-	3148-	4168-

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

Rated voltage (+85°C) 1/ V dc	Cap. µF	Cap. tolerance percent	DC leakage (max) +25°C µA	DC leakage (max) +85°C, +125°C µA	Dissipation factor (max) percent	ESR at +25°C (max) 2/ Ohms	Derated voltage (+125°C) 1/ V dc	Surge voltage (+85°C) V dc	Impedance (max) Ohms	Capacitance change at -55°C percent	Capacitance change at +85°C percent	Capacitance change at +125°C percent	Max 85°C 40 kHz ripple current 3/ mA rms	Case size	PIN M39006/30- 4/ 5/ noncompression seal Failure rate level (%/1,000 hour) M(1.0)	P(0.1)	R(0.01)	S(0.001)	PIN M39006/30- 4/ 5/ compression seal Failure rate level (%/1,000 hour) M(1.0)	P(0.1)	R(0.01)	S(0.001)
30	40	±5	1	5	5	1.66	20	34.5	65	-24	+10.5	+12	1,120	T2	0109-	0329-	0549-	0769-	1109-	2119-	3149-	4169-
30	68	±20	1	8	6.5	1.27	20	34.5	60	-24	+13	+15	1,285	T2	0110-	0330-	0550-	0770-	1110-	2120-	3150-	4170-
30	68	±10	1	8	6.5	1.27	20	34.5	60	-24	+13	+15	1,285	T2	0111-	0331-	0551-	0771-	1111-	2121-	3151-	4171-
30	68	±5	1	8	6.5	1.27	20	34.5	60	-24	+13	+15	1,285	T2	0112-	0332-	0552-	0772-	1112-	2122-	3152-	4172-
30	100	±20	2	12	8.5	1.13	20	34.5	40	-28	+10.5	+12	1,450	T3	0113-	0333-	0553-	0773-	1113-	2123-	3153-	4173-
30	100	±10	2	12	8.5	1.13	20	34.5	40	-28	+10.5	+12	1,450	T3	0114-	0334-	0554-	0774-	1114-	2124-	3154-	4174-
30	100	±5	2	12	8.5	1.13	20	34.5	40	-28	+10.5	+12	1,450	T3	0115-	0335-	0555-	0775-	1115-	2125-	3155-	4175-
30	150	±20	2	18	11.5	1.02	20	34.5	35	-48	+13	+15	1,525	T3	0116-	0336-	0556-	0776-	1116-	2126-	3156-	4176-
30	150	±10	2	18	11.5	1.02	20	34.5	35	-48	+13	+15	1,525	T3	0117-	0337-	0557-	0777-	1117-	2127-	3157-	4177-
30	150	±5	2	18	11.5	1.02	20	34.5	35	-48	+13	+15	1,525	T3	0118-	0338-	0558-	0778-	1118-	2128-	3158-	4178-
30	300	±20	8	32	15.5	0.69	20	34.5	25	-60	+25	+25	1,950	T4	0119-	0339-	0559-	0779-	1119-	2129-	3159-	4179-
30	300	±10	8	32	15.5	0.69	20	34.5	25	-60	+25	+25	1,950	T4	0120-	0340-	0560-	0780-	1120-	2130-	3160-	4180-
50	5	±20	1	2	1.5	3.98	30	57.5	400	-16	+ 5	+ 6	580	T1	0121-	0341-	0561-	0781-	1121-	2131-	3161-	4181-
50	5	±10	1	2	1.5	3.98	30	57.5	400	-16	+ 5	+ 6	580	T1	0122-	0342-	0562-	0782-	1122-	2132-	3162-	4182-
50	5	±5	1	2	1.5	3.98	30	57.5	400	-16	+ 5	+ 6	580	T1	0123-	0343-	0563-	0783-	1123-	2133-	3163-	4183-
50	10	±20	1	2	2	2.66	30	57.5	250	-24	+ 8	+ 9	715	T1	0124-	0344-	0564-	0784-	1124-	2134-	3164-	4184-
50	10	±10	1	2	2	2.66	30	57.5	250	-24	+ 8	+ 9	715	T1	0125-	0345-	0565-	0785-	1125-	2135-	3165-	4185-
50	10	±5	1	2	2	2.66	30	57.5	250	-24	+ 8	+ 9	715	T1	0126-	0346-	0566-	0786-	1126-	2136-	3166-	4186-
50	25	±20	1	5	4	2.13	30	57.5	95	-20	+10.5	+12	1,005	T2	0127-	0347-	0567-	0787-	1127-	2137-	3167-	4187-
50	25	±10	1	5	4	2.13	30	57.5	95	-20	+10.5	+12	1,005	T2	0128-	0348-	0568-	0788-	1128-	2138-	3168-	4188-
50	25	±5	1	5	4	2.13	30	57.5	95	-20	+10.5	+12	1,005	T2	0129-	0349-	0569-	0789-	1129-	2139-	3169-	4189-
50	47	±20	1	9	5.5	1.56	30	57.5	70	-28	+13	+15	1,155	T2	0130-	0350-	0570-	0790-	1130-	2140-	3170-	4190-
50	47	±10	1	9	5.5	1.56	30	57.5	70	-28	+13	+15	1,155	T2	0131-	0351-	0571-	0791-	1131-	2141-	3171-	4191-
50	47	±5	1	9	5.5	1.56	30	57.5	70	-28	+13	+15	1,155	T2	0132-	0352-	0572-	0792-	1132-	2142-	3172-	4192-
50	60	±20	2	12	6	1.33	30	57.5	45	-16	+10.5	+12	1,335	T3	0133-	0353-	0573-	0793-	1133-	2143-	3173-	4193-
50	60	±10	2	12	6	1.33	30	57.5	45	-16	+10.5	+12	1,335	T3	0134-	0354-	0574-	0794-	1134-	2144-	3174-	4194-
50	60	±5	2	12	6	1.33	30	57.5	45	-16	+10.5	+12	1,335	T3	0135-	0355-	0575-	0795-	1135-	2145-	3175-	4195-
50	82	±20	2	16	7.5	1.22	30	57.5	45	-32	+13	+15	1,400	T3	0136-	0356-	0576-	0796-	1136-	2146-	3176-	4196-
50	82	±10	2	16	7.5	1.22	30	57.5	45	-32	+13	+15	1,400	T3	0137-	0357-	0577-	0797-	1137-	2147-	3177-	4197-
50	82	±5	2	16	7.5	1.22	30	57.5	45	-32	+13	+15	1,400	T3	0138-	0358-	0578-	0798-	1138-	2148-	3178-	4198-
50	160	±20	8	32	8.5	0.71	30	57.5	27	-50	+25	+25	1,900	T4	0139-	0359-	0579-	0799-	1139-	2149-	3179-	4199-
50	160	±10	8	32	8.5	0.71	30	57.5	27	-50	+25	+25	1,900	T4	0140-	0360-	0580-	0800-	1140-	2150-	3180-	4200-
60	4	±20	1	2	1.4	4.65	40	69.0	550	-16	+ 5	+ 6	525	T1	0141-	0361-	0581-	0801-	1141-	2151-	3181-	4201-
60	4	±10	1	2	1.4	4.65	40	69.0	550	-16	+ 5	+ 6	525	T1	0142-	0362-	0582-	0802-	1142-	2152-	3182-	4202-
60	4	±5	1	2	1.4	4.65	40	69.0	550	-16	+ 5	+ 6	525	T1	0143-	0363-	0583-	0803-	1143-	2153-	3183-	4203-
60	8.2	±20	1	2	2	3.24	40	69.0	275	-24	+ 8	+ 9	625	T1	0144-	0364-	0584-	0804-	1144-	2154-	3184-	4204-

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

Rated voltage (+85°C) 1/ V dc	Cap. µF	Cap. tolerance percent	DC leakage (max) +25°C µA	DC leakage (max) +85°C, +125°C µA	Dissipation factor (max) percent	ESR at +25°C (max) 2/ Ohms	Derated voltage (+125°C) 1/ V dc	Surge voltage (+85°C) V dc	Impedance (max) Ohms	Capacitance Change at -55°C percent	Capacitance Change at +85°C percent	Capacitance Change at +125°C percent	Max 85°C 40 kHz ripple current 3/ mA rms	Case size	PIN M39006/30- 4/ 5/ noncompression seal Failure rate level (%/1,000 hour) M(1.0)	P(0.1)	R(0.01)	S(0.001)	PIN M39006/30- 4/ 5/ compression seal Failure rate level (%/1,000 hour) M(1.0)	P(0.1)	R(0.01)	S(0.001)
60	8.2	±10	1	2	2	3.24	40	69.0	275	-24	+ 8	+ 9	625	T1	0145-	0365-	0585-	0805-	1145-	2155-	3185-	4205-
60	8.2	± 5	1	2	2	3.24	40	69.0	275	-24	+ 8	+ 9	625	T1	0146-	0366-	0586-	0806-	1146-	2156-	3186-	4206-
60	20	±20	1	5	3.5	2.32	40	69.0	105	-16	+10.5	+12	930	T2	0147-	0367-	0587-	0807-	1147-	2157-	3187-	4207-
60	20	±10	1	5	3.5	2.32	40	69.0	105	-16	+10.5	+12	930	T2	0148-	0368-	0588-	0808-	1148-	2158-	3188-	4208-
60	20	± 5	1	5	3.5	2.32	40	69.0	105	-16	+10.5	+12	930	T2	0149-	0369-	0589-	0809-	1149-	2159-	3189-	4209-
60	39	±20	1	9	5	1.70	40	69.0	90	-28	+10.5	+12	1,110	T2	0150-	0370-	0590-	0810-	1150-	2160-	3190-	4210-
60	39	±10	1	9	5	1.70	40	69.0	90	-28	+10.5	+12	1,110	T2	0151-	0371-	0591-	0811-	1151-	2161-	3191-	4211-
60	39	± 5	1	9	5	1.70	40	69.0	90	-28	+10.5	+12	1,110	T2	0152-	0372-	0592-	0812-	1152-	2162-	3192-	4212-
60	50	±20	2	12	5	1.33	40	69.0	50	-16	+10.5	+12	1,330	T3	0153-	0373-	0593-	0813-	1153-	2163-	3193-	4213-
60	50	±10	2	12	5	1.33	40	69.0	50	-16	+10.5	+12	1,330	T3	0154-	0374-	0594-	0814-	1154-	2164-	3194-	4214-
60	50	± 5	2	12	5	1.33	40	69.0	50	-16	+10.5	+12	1,330	T3	0155-	0375-	0595-	0815-	1155-	2165-	3195-	4215-
60	68	±20	2	16	6.5	1.27	40	69.0	50	-32	+10.5	+12	1,365	T3	0156-	0376-	0596-	0816-	1156-	2166-	3196-	4216-
60	68	±10	2	16	6.5	1.27	40	69.0	50	-32	+10.5	+12	1,365	T3	0157-	0377-	0597-	0817-	1157-	2167-	3197-	4217-
60	68	± 5	2	16	6.5	1.27	40	69.0	50	-32	+10.5	+12	1,365	T3	0158-	0378-	0598-	0818-	1158-	2168-	3198-	4218-
60	140	±20	8	32	8	0.76	40	69.0	28	-40	+20	+20	1,850	T4	0159-	0379-	0599-	0819-	1159-	2169-	3199-	4219-
60	140	±10	8	32	8	0.76	40	69.0	28	-40	+20	+20	1,850	T4	0160-	0380-	0600-	0820-	1160-	2170-	3200-	4220-
75	3.5	±20	1	2	1.25	4.74	50	86.2	650	-16	+ 5	+ 6	525	T1	0161-	0381-	0601-	0821-	1161-	2171-	3201-	4221-
75	3.5	±10	1	2	1.25	4.74	50	86.2	650	-16	+ 5	+ 6	525	T1	0162-	0382-	0602-	0822-	1162-	2172-	3202-	4222-
75	3.5	± 5	1	2	1.25	4.74	50	86.2	650	-16	+ 5	+ 6	525	T1	0163-	0383-	0603-	0823-	1163-	2173-	3203-	4223-
75	6.8	±20	1	2	1.75	3.42	50	86.2	300	-20	+ 8	+ 9	610	T1	0164-	0384-	0604-	0824-	1164-	2174-	3204-	4224-
75	6.8	±10	1	2	1.75	3.42	50	86.2	300	-20	+ 8	+ 9	610	T1	0165-	0385-	0605-	0825-	1165-	2175-	3205-	4225-
75	6.8	± 5	1	2	1.75	3.42	50	86.2	300	-20	+ 8	+ 9	610	T1	0166-	0386-	0606-	0826-	1166-	2176-	3206-	4226-
75	15	±20	1	5	3	2.66	50	86.2	150	-16	+ 8	+ 9	890	T2	0167-	0387-	0607-	0827-	1167-	2177-	3207-	4227-
75	15	±10	1	5	3	2.66	50	86.2	150	-16	+ 8	+ 9	890	T2	0168-	0388-	0608-	0828-	1168-	2178-	3208-	4228-
75	15	± 5	1	5	3	2.66	50	86.2	150	-16	+ 8	+ 9	890	T2	0169-	0389-	0609-	0829-	1169-	2179-	3209-	4229-
75	33	±20	1	10	5	2.01	50	86.2	90	-24	+10.5	+15	1,000	T2	0170-	0390-	0610-	0830-	1170-	2180-	3210-	4230-
75	33	±10	1	10	5	2.01	50	86.2	90	-24	+10.5	+15	1,000	T2	0171-	0391-	0611-	0831-	1171-	2181-	3211-	4231-
75	33	± 5	1	10	5	2.01	50	86.2	90	-24	+10.5	+15	1,000	T2	0172-	0392-	0612-	0832-	1172-	2182-	3212-	4232-
75	40	±20	2	12	4.5	1.50	50	86.2	60	-16	+10.5	+12	1,250	T3	0173-	0393-	0613-	0833-	1173-	2183-	3213-	4233-
75	40	±10	2	12	4.5	1.50	50	86.2	60	-16	+10.5	+12	1,250	T3	0174-	0394-	0614-	0834-	1174-	2184-	3214-	4234-
75	40	± 5	2	12	4.5	1.50	50	86.2	60	-16	+10.5	+12	1,250	T3	0175-	0395-	0615-	0835-	1175-	2185-	3215-	4235-
75	56	±20	2	17	5.5	1.31	50	86.2	60	-28	+10.5	+15	1,335	T3	0176-	0396-	0616-	0836-	1176-	2186-	3216-	4236-
75	56	±10	2	17	5.5	1.31	50	86.2	60	-28	+10.5	+15	1,335	T3	0177-	0397-	0617-	0837-	1177-	2187-	3217-	4237-
75	56	± 5	2	17	5.5	1.31	50	86.2	60	-28	+10.5	+15	1,335	T3	0178-	0398-	0618-	0838-	1178-	2188-	3218-	4238-
75	110	±20	9	36	6	0.73	50	86.2	29	-35	+20	+20	1,850	T4	0179-	0399-	0619-	0839-	1179-	2189-	3219-	4239-
75	110	±10	9	36	6	0.73	50	86.2	29	-35	+20	+20	1,850	T4	0180-	0400-	0620-	0840-	1180-	2190-	3220-	4240-

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

Rated voltage (+85°C) 1/ V dc	Cap. µF	Cap. tolerance percent	DC leakage (max) +25°C µA	DC leakage (max) +85°C, +125°C µA	Dissipation factor (max) percent	ESR at +25°C (max) 2/ Ohms	Derated voltage (+125°C) 1/ V dc	Surge voltage (+85°C) V dc	Impedance (max) Ohms	Capacitance change at -55°C percent	Capacitance change at +85°C percent	Capacitance change at +125°C percent	Max 85°C 40 kHz ripple current 3/ mA rms	Case size	PIN M39006/30- 4/ 5/ noncompression seal Failure rate level (%/1,000 hour) M(1.0)	P(0.1)	R(0.01)	S(0.001)	PIN M39006/30- 4/ 5/ compression seal Failure rate level (%/1,000 hour) M(1.0)	P(0.1)	R(0.01)	S(0.001)
100	2.5	±20	1	2	1	5.31	65	115	950	-16	+ 7	+ 8	505	T1	0181-	0401-	0621-	0841-	1181-	2191-	3221-	4241-
100	2.5	±10	1	2	1	5.31	65	115	950	-16	+ 7	+ 8	505	T1	0182-	0402-	0622-	0842-	1182-	2192-	3222-	4242-
100	2.5	± 5	1	2	1	5.31	65	115	950	-16	+ 7	+ 8	505	T1	0183-	0403-	0623-	0843-	1183-	2193-	3223-	4243-
100	4.7	±20	1	2	1.5	4.24	65	115	500	-16	+ 7	+ 8	565	T1	0184-	0404-	0624-	0844-	1184-	2194-	3224-	4244-
100	4.7	±10	1	2	1.5	4.24	65	115	500	-16	+ 7	+ 8	565	T1	0185-	0405-	0625-	0845-	1185-	2195-	3225-	4245-
100	4.7	± 5	1	2	1.5	4.24	65	115	500	-16	+ 7	+ 8	565	T1	0186-	0406-	0626-	0846-	1186-	2196-	3226-	4246-
100	11	±20	1	4	2.5	3.02	65	115	200	-16	+ 8	+ 8	835	T2	0187-	0407-	0627-	0847-	1187-	2197-	3227-	4247-
100	11	±10	1	4	2.5	3.02	65	115	200	-16	+ 8	+ 8	835	T2	0188-	0408-	0628-	0848-	1188-	2198-	3228-	4248-
100	11	± 5	1	4	2.5	3.02	65	115	200	-16	+ 8	+ 8	835	T2	0189-	0409-	0629-	0849-	1189-	2199-	3229-	4249-
100	22	±20	1	9	3.75	2.26	65	115	100	-16	+ 8	+ 8	965	T2	0190-	0410-	0630-	0850-	1190-	2200-	3230-	4250-
100	22	±10	1	9	3.75	2.26	65	115	100	-16	+ 8	+ 8	965	T2	0191-	0411-	0631-	0851-	1191-	2201-	3231-	4251-
100	22	± 5	1	9	3.75	2.26	65	115	100	-16	+ 8	+ 8	965	T2	0192-	0412-	0632-	0852-	1192-	2202-	3232-	4252-
100	30	±20	2	12	3.5	1.55	65	115	80	-16	+ 8	+ 8	1,240	T3	0193-	0413-	0633-	0853-	1193-	2203-	3233-	4253-
100	30	±10	2	12	3.5	1.55	65	115	80	-16	+ 8	+ 8	1,240	T3	0194-	0414-	0634-	0854-	1194-	2204-	3234-	4254-
100	30	± 5	2	12	3.5	1.55	65	115	80	-16	+ 8	+ 8	1,240	T3	0195-	0415-	0635-	0855-	1195-	2205-	3235-	4255-
100	43	±20	2	17	4.25	1.31	65	115	70	-20	+ 8	+ 8	1,335	T3	0196-	0416-	0636-	0856-	1196-	2206-	3236-	4256-
100	43	±10	2	17	4.25	1.31	65	115	70	-20	+ 8	+ 8	1,335	T3	0197-	0417-	0637-	0857-	1197-	2207-	3237-	4257-
100	43	± 5	2	17	4.25	1.31	65	115	70	-20	+ 8	+ 8	1,335	T3	0198-	0418-	0638-	0858-	1198-	2208-	3238-	4258-
100	86	±20	9	36	5	0.77	65	115	30	-25	+15	+15	1,800	T4	0199-	0419-	0639-	0859-	1199-	2209-	3239-	4259-
100	86	±10	9	36	5	0.77	65	115	30	-25	+15	+15	1,800	T4	0200-	0420-	0640-	0860-	1200-	2210-	3240-	4260-
125	1.7	±20	1	2	1	7.81	85	144	1,250	-16	+ 7	+ 8	415	T1	0201-	0421-	0641-	0861-	1201-	2211-	3241-	4261-
125	1.7	±10	1	2	1	7.81	85	144	1,250	-16	+ 7	+ 8	415	T1	0202-	0422-	0642-	0862-	1202-	2212-	3242-	4262-
125	1.7	± 5	1	2	1	7.81	85	144	1,250	-16	+ 7	+ 8	415	T1	0203-	0423-	0643-	0863-	1203-	2213-	3243-	4263-
125	3.6	±20	1	2	1.35	4.98	85	144	600	-16	+ 7	+ 8	520	T1	0204-	0424-	0644-	0864-	1204-	2214-	3244-	4264-
125	3.6	±10	1	2	1.35	4.98	85	144	600	-16	+ 7	+ 8	520	T1	0205-	0425-	0645-	0865-	1205-	2215-	3245-	4265-
125	3.6	± 5	1	2	1.35	4.98	85	144	600	-16	+ 7	+ 8	520	T1	0206-	0426-	0646-	0866-	1206-	2216-	3246-	4266-
125	9	±20	1	5	2.5	3.69	85	144	240	-16	+ 7	+ 8	755	T2	0207-	0427-	0647-	0867-	1207-	2217-	3247-	4267-
125	9	±10	1	5	2.5	3.69	85	144	240	-16	+ 7	+ 8	755	T2	0208-	0428-	0648-	0868-	1208-	2218-	3248-	4268-
125	9	± 5	1	5	2.5	3.69	85	144	240	-16	+ 7	+ 8	755	T2	0209-	0429-	0649-	0869-	1209-	2219-	3249-	4269-
125	14	±20	1	7	3	2.85	85	144	167	-16	+ 7	+ 8	860	T2	0210-	0430-	0650-	0870-	1210-	2220-	3250-	4270-
125	14	±10	1	7	3	2.85	85	144	167	-16	+ 7	+ 8	860	T2	0211-	0431-	0651-	0871-	1211-	2221-	3251-	4271-
125	14	± 5	1	7	3	2.85	85	144	167	-16	+ 7	+ 8	860	T2	0212-	0432-	0652-	0872-	1212-	2222-	3252-	4272-

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

Rated voltage (+85°C) 1/ V dc	Cap. µF	Cap. tolerance percent	DC leakage (max) +25°C µA	DC leakage (max) +85°C, +125°C µA	Dissipation factor (max) percent	ESR at +25°C (max) 2/ Ohms	Derated voltage (+125°C) 1/ V dc	Surge voltage (+85°C) V dc	Impedance (max) Ohms	Capacitance change at -55°C percent	Capacitance change at +85°C percent	Capacitance change at +125°C percent	Max 85°C 40 kHz ripple current 3/ mA rms	Case size	PIN M39006/30- 4/ 5/ noncompression seal Failure rate level (%/1,000 hour) M(1.0)	P(0.1)	R(0.01)	S(0.001)	PIN M39006/30- 4/ 5/ compression seal Failure rate level (%/1,000 hour) M(1.0)	P(0.1)	R(0.01)	S(0.001)
125	18	±20	2	9	2.5	1.85	85	144	129	-16	+ 7	+ 8	1,130	T3	0213-	0433-	0653-	0873-	1213-	2223-	3253-	4273-
125	18	±10	2	9	2.5	1.85	85	144	129	-16	+ 7	+ 8	1,130	T3	0214-	0434-	0654-	0874-	1214-	2224-	3254-	4274-
125	18	± 5	2	9	2.5	1.85	85	144	129	-16	+ 7	+ 8	1,130	T3	0215-	0435-	0655-	0875-	1215-	2225-	3255-	4275-
125	25	±20	2	13	3	1.59	85	144	93	-16	+ 7	+ 8	1,200	T3	0216-	0436-	0656-	0876-	1216-	2226-	3256-	4276-
125	25	±10	2	13	3	1.59	85	144	93	-16	+ 7	+ 8	1,200	T3	0217-	0437-	0657-	0877-	1217-	2227-	3257-	4277-
125	25	± 5	2	13	3	1.59	85	144	93	-16	+ 7	+ 8	1,200	T3	0218-	0438-	0658-	0878-	1218-	2228-	3258-	4278-
125	56	±20	10	40	3.25	0.77	85	144	32	-25	+15	+15	1,800	T4	0219-	0439-	0659-	0879-	1219-	2229-	3259-	4279-
125	56	±10	10	40	3.25	0.77	85	144	32	-25	+15	+15	1,800	T4	0220-	0440-	0660-	0880-	1220-	2230-	3260-	4280-

1/ Reverse voltage rating at 85°C is 3 V dc and at 125°C is 2 V dc.

2/ Maximum ESR is calculated by the following equation:

$$\text{ESR (max)} = \frac{DF}{2\pi f C}, \text{ where:}$$

DF = Maximum dissipation factor from table 1;
f = 120 Hz;
C = Nominal capacitance.

3/ For ripple current limits at various temperatures, voltages, and frequencies, see table II.

* 4/ Dash number shall include the letter “H” to indicate the optional vibration and shock requirements (i.e., 53.79 g’s random vibration, 80 g’s sinusoidal vibration, and 500 g’s shock) or the “-“ will be deleted. The acquiring activity may substitute an “H” designated part in place of a non “H” designated part at their option.

5/ Compression seal is generally referred to as single-seal construction; noncompression seal is generally referred to as double-seal construction.

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TABLE II. Ripple current multipliers versus frequency, temperature, and applied peak voltage. 1/ 2/ 3/ 4/ 5/

Frequency of applied ripple current		120 Hz				800 Hz				1 kHz				10 kHz				40 kHz				100 kHz			
		Temp °C				Temp °C				Temp °C				Temp °C				Temp °C				Temp °C			
	Ambient still air	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°	≤55°	85°	105°	125°
	100 percent	.60	.39	---	---	.71	.43	---	---	.72	.45	---	---	.88	.55	---	---	1.0	.63	---	---	1.1	.69	---	---
	90 percent	.60	.46	---	---	.71	.55	---	---	.72	.55	---	---	.88	.67	---	---	1.0	.77	---	---	1.1	.85	---	---
6/	80 percent	.60	.52	.35	---	.71	.62	.42	---	.72	.62	.42	---	.88	.76	.52	---	1.0	.87	.59	---	1.1	.96	.65	---
	70 percent	.60	.58	.44	---	.71	.69	.52	---	.72	.70	.52	---	.88	.85	.64	---	1.0	.97	.73	---	1.1	1.07	.80	---
	≤ 66 2/3 percent	.60	.60	.46	.27	.71	.71	.55	.32	.72	.72	.55	.32	.88	.88	.68	.40	1.0	1.0	.77	.45	1.1	1.1	.85	.50

1/ At 125°C, the rated voltage of the capacitors decreases to 66.67 percent of the 85°C rated voltage.

2/ The peak of the applied ac ripple voltage plus the applied dc voltage must not exceed the dc voltage rating of the capacitor either forward or reverse.

3/ The ripple current listed in table I represents a rating calculated by using a maximum internal temperature rise (ΔT) at 50°C at 40 kHz at 85°C ambient temperature, with a maximum peak rated voltage of 66.67 percent of the 85°C peak voltage rating.

4/ The maximum allowable internal temperature rise (ΔT) decreases linearly to a calculated 10°C rise at 125°C ambient.

5/ The internal temperature rise is directly proportional to the ESR of the capacitor, and ESR increases with decreasing frequency.

6/ Percentage of 85°C rated peak voltage.

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Rated voltage: See table I.

Capacitance (Cap.) nominal (nom): See table I.

Cap. tolerance: See table I.

Failure rate level: M, P, R, and S (see table I).

DC leakage (DCL) (max) (at +25°C): See table I.

Cap. (nominal): Method 305 of MIL-STD-202 (see table I).

Dissipation factor (DF) (max): See table I.

Seal: Method 112 of MIL-STD-202, conditions A or D, and C.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100 g's) or condition D (500 g's), as applicable.

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 g's) or condition H (80 g's), as applicable.

Random vibration (when applicable): Method 214 of MIL-STD-202, test condition II-K (51 g's).

Thermal shock: Method 107 of MIL-STD-202, condition A (with step 3 at +125°C).

* Number of cycles: In accordance with MIL-PRF-39006, the following shall apply.

DCL: 200 percent (max) of +25°C value of table I for qualification and group C.

DCL: 125 percent (max) of +25°C value of table I for group B.

ΔCap: Within ±5 percent of initial measurement.

DF: 115 percent (max) of initial requirement.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).

Solderability: Method 208 of MIL-STD-202.

Terminal strength:

Pull: Method 211 of MIL-STD-202, condition A.

Wire-lead bend: In accordance with MIL-PRF-39006.

Surge voltage: See table I.

Moisture resistance: Method 106 of MIL-STD-202.

DCL: 125 percent (max) of +25°C value of table I.

ΔCap: Within ±8 percent of initial measurement.

DF: 115 percent (max) of initial requirement.

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Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202. 2,000 V dc, minimum.

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 100 megohms, minimum.

Low temperature (storage): Method 502 of MIL-STD-810.

DCL: See table I.

ΔCap: Within ±5 percent of initial measurement.

DF: See table I.

Stability at low and high temperatures:

Step 1 (+25°C):

DCL: See table I.

ΔCap: Within tolerance of table I value.

DF: See table I.

Step 2 (-55°C):

Impedance. See table I.

ΔCap: See table I.

Step 3 (+25°C):

DCL: See table I.

ΔCap: Within ±5 percent of step 1 value.

DF: See table I.

Step 4 (+85°C):

DCL: See table I.

ΔCap: See table I.

DF: See table I.

Step 5 (+125°C):

DCL: See table I.

ΔCap: See table I.

DF: See table I.

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Step 6 (+25°C):

DCL: See table I.

ΔCap: Within ±5 percent of step 1 value.

DF: See table I.

Reverse voltage: As specified in MIL-PRF-39006.

DCL: See table I.

ΔCap: Within tolerance of table I value.

DF: See table I.

Life at 85°C: Method 108 of MIL-STD-202.

2,000-hour (qualification):

DCL: See table I.

ΔCap: Within ± 10 percent of initial measurement.

DF: See table I.

10,000-hour (group B) and extended life:

DCL (85°C): 125 percent (max) of table I value.

DCL (25°C): See table I.

ΔCap: Within +10 percent, -20 percent of initial measurement.

DF: 200 percent (max) of table I value.

DWV: 2,000 V dc, minimum.

IR.: 100 megohms, minimum.

AC ripple life at 85°C (qualification and group C):

Test ripple current: See table I.

Test voltage: 66.67 percent of rated voltage (see table I).

DCL: See table I.

ΔCap: Within ±10 percent of initial measurement.

DF: See table I.

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Life at 125°C: Method 108 of MIL-STD-202.

DCL: See table I.

ΔCap: Within ± 10 percent of initial measurement.

DF: See table I.

DWV: 2,000 V dc, minimum.

IR.: 100 megohms, minimum.

2,000-hour (group C):

DCL (125°C): 125 percent (max) of table I value.

DCL (25°C): See table I.

ΔCap: Within +10 percent, -20 percent of initial measurement.

DF: 200 percent (max) of table I value.

DWV: 2,000 V dc, minimum.

IR.: 100 megohms, minimum.

Barometric pressure: Method 105 of MIL-STD-202, condition E (150,000 feet) (45,720.1 m).

Test voltage: Rated voltage.

* Resistance to soldering heat: Method 210 of MIL-STD-202, condition C.

DCL: See table I.

ΔCap: Within ±5 percent of initial measurement.

DF: See table I.

Marking: In accordance with MIL-PRF-39006, with the following exception: The manufacturer may mark the T1 case size "H" designated capacitors as follows:

Example: M39006/
30H0001
JAN TM
12345
9415A

Polarity of all parts: Series of (+) symbols encircling positive end (see figure 1).

Voltage groups (85°C rated voltage):

I - - - - - - - - - - 6 V through 15 V

II - - - - - - - - - - 25 V through 75 V.

III - - - - - - - - - - 100 V and 125 V.

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* Substitution of H designated parts for non H designated parts: The acquiring activity may substitute an H designated part in place of a non H designated part at their option.

* CHANGES FROM PREVIOUS ISSUE: The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* Referenced documents. In addition to MIL-PRF-39006, this document also references MIL-STD-202, MIL-PRF-39006/22, MIL-STD-810 and MIL-STD-1276.

Custodians:
Army - CR
Navy - EC
Air Force – 11
DLA - CC

Review activities:
Navy - AS, MC, OS
Air Force – 19

Preparing activity:
DLA - CC

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